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(1) Research Program/Project/Study Title: Darag Native Chicken, Bamboo Shoot and Malunggay Leaves Siomai	
Proponent Name:	Dr. Maryneth B. Barrios
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Dumarao
		Province:	Capiz
		Region:	6
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	

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(7) Sector/Commodity Livestock and Crop	(8) Discipline/Field of Specialization Foods
(8) Rationale Serious food travelers know that a lot of the best food finds can be found not in the fanciest establishments but on the streets. Street food has become an important food category on its own, especially when it comes to Filipino food. Street food can still be found on almost every street corner in the major cities throughout the Philippines. It is a testament to the adaptability and ingenuity of the Filipino people that these parts have been turned into some of the most delicious and most affordable bites of food (Tee, 2021). Among of these street foods are siomai or dumplings. Siomai is a popular Filipino dish that is made up of a seasoned mixture of ground pork, shrimp, and vegetables, all wrapped in a thin wonton wrapper. It is typically served with a spicy soy-based dipping sauce, and is a popular snack food in the Philippines. It is often served as an appetizer or as a main dish with rice and other accompaniments. Siomai is also a popular street food in the Philippines, and can be found in almost any market or street vendor. It is a versatile dish, and can be enjoyed as a snack, appetizer, or main dish. Siomai is believed to have originated in China as early as the 10th century and was introduced to Philippines which became the all time favorite of many Filipinos (Martin, 2023). Since siomai are considered “junk food” or pejorative term for food containing high levels of calories from sugar or fat with little protein, vitamins or minerals, many studies and researches were conducted to look for high nutritious materials to serve as raw materials for the production of siomai. This is the use of some healthy plants and meat. One of these plants is bamboo shoots of <i>Gigantochloa levis</i> (Botong) bamboo variety because it is a good source of fiber, potassium, antioxidants, and lower in fat and calories. Darag native chicken for meat substitute because of its tasty and lean meat and at the same time lower in cholesterol. On the other hand, malunggay (<i>Moringa oleifera</i>) is recognized as an excellent source of phytochemicals, with potential applications in functional and medicinal food preparations due to its nutritional and medicinal properties. For centuries, Malunggay has been used as a system of Ayurvedic and Unani medicine and has a wide range of nutritional and bioactive compounds, including proteins, essential amino acids, carbohydrates, lipids, fibre, vitamins, minerals, phenolic compounds, phytosterols and others (Milla et al., 2021). This study will utilized the products of the two established research of CAPSU DUMARAO – the Bamboo Research Project and Darag Native Chicken Project. Bamboo shoots from Bamboo Research Project and Growers for slaughter from Darag native chicken project will be used.	
(9-A) Project Description This study will be another concept of the researcher in continuation of DABAM siomai and shanghai study. This will utilized the available resources that is found in the campus, the Darag native chicken and Bamboo and Malunggay leaves. This will used slaughter Darag native chickens and Bamboo shoots. The first step of the study will be the evaluation of what combination (ratio) of the Darag native chicken, bamboo shoot and malunggay leaves is more acceptable in the taster. This will followed by the evaluation of the shelf life and the nutritional value. Further more, the product from this study will also applied for utility Model certificate in the IPOPHIL and will be introduce in the market specifically in the school canteen.	
(9) Objectives/Statement of the Problem 1. To determine the quality of Darag native chicken, bamboo shoots and Malunggay leaves siomai in terms of taste, aroma, and texture. 2. To find out which of Darag native chicken meat, bamboo shoot and malunggay leaves combination has the best acceptability level. 3. To determine the shelf life of the siomai.	

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4. To evaluate the nutritional value of the siomai.

(11) Review of Literature/Prior Art Search

Darag native chicken are indigenous strains which are located only in Western Visayas. (WESVARDEC,2009). They are well known for their adaptability to local agro climatic conditions,hardiness, ability to utilize farm by products, tasty meat and a resistance to disease (PCAARD, 2007).One of the reasons that the native chicken are regaining their popularity in the Philippines is the fact that they have a unique, and arguably better, taste as compared to the commercial varieties. That applies to both their meat and their eggs (<http://poultrymanual.com>). Darag Chicken meat, when cooked, contains less fat, is high protein, rich in potassium and has more unsaturated fatty acids.Famous local Darag delicacies are tinola, litson manok, tinu-om and chicken inasal (Fernandez, 2014).

Bamboo shoots is a good source of fiber, potassium, anti oxidants, and lower in fats and calories. The juvenile shoots are not only delicious but are rich in nutrient components,mainly proteins, carbohydrates,minerals, and fiber and are low in fat and sugars. In addition, they contain phytosterols and a high amount of fiber that can be labeled as a nutraceuticals or natural medicine that are attracting the attention of health advocates and scientist alike. The shoots are free from residual toxicity and grow without the application of fertilizers (Chongtam et al., 2011).

Moringa oleifera belongs to the Moringaceae family and is the best known of the native *Moringa oleifera* genus. For centuries, it has been used as a system of Ayurvedic and Unani medicine and has a wide range of nutritional and bioactive compounds, including proteins, essential amino acids, carbohydrates, lipids, fibre, vitamins, minerals, phenolic compounds, phytosterols and others. These characteristics allow it to have pharmacological properties, including anti-diabetic, anti-inflammatory, anticarcinogenic, antioxidant, cardioprotective, antimicrobial and hepatoprotective properties. The entire *Moringa oleifera* plant is edible, including its flowers, however, it is not entirely safe, because of compounds that have been found mainly in the root and bark, so the leaf was identified as the safest. *Moringa oleifera* is recognised as an excellent source of phytochemicals, with potential applications in functional and medicinal food preparations due to its nutritional and medicinal properties (Milla et al., 2021).

The technology produced from this study relates to registered Utility Model PH2/2018/050508 - Composition of Bamboo Shoot Siomai of Rhodora Ticar, Jhunel Luna and Erika Alayon of CapSU Mambusao (Patentscope, 2024) in terms of some of its ingredients and process of preparation, however, the technology is apparently different in its major ingredients which is the Darag native chicken and malunggay leaves.

(12) Scientific Basis/Theoretical Framework

Aliperio et al., (2017) studies the acceptability level of bamboo shoots in making siomai and the result of their study showed that bamboo shoots can be used as substitute in making siomai. Furthermore, their study revealed that the quality of siomai has a significant differences in terms of taste, aroma and texture.

Barrios et al., (2019) studies showed that 25% to 75% mixture of Darag native chicken and Bamboo shoot siomai was found satisfactory to very satisfactory acceptability level in terms of taste, aroma and texture.

(13) Methodology

The materials used in the study will includes chopping board, knife, pan, mixing bowl, mortar and pestle, wood ladle, and whisk. The ingredients will be the *Gigatochloa levis* (botong) shoots, Darag native chicken ground meat, Malunggay leaves, ground pepper, seasoning, garlic, onions, carrots, egg, water, cooking oil and all purpose flour.

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Quality Assessment of Siomai

To determine the quality of Siomai, five treatments will be utilized in the study and this will be arranged in Completely Randomized Design (CRD). Each treatment will be replicated three times.

This treatment will be:

T1 - 100% bamboo shoot

T2 - Darag native chicken meat + bamboo shoot

T3 - Darag native chicken meat + malunggay leaves

T4 - Darag native chicken meat + bamboo shoot + malunggay leaves

T5 - 100% Darag native chicken meat

This Darag native chicken will be taken from the Darag Native Chicken Project of CapSU Dumarao. The growers chicken aging 4 months and above will be used in the study. This will be slaughtered and the meat will be grounded. The bamboo shoots will also be taken from the Bamboo Research Project of the campus. Young shoots will be utilized. This will be cleaned and shredded. While Malunggay leaves will be collected also at the Darag Research site. Fresh leaves will be chopped and shredded.

To evaluate the quality of the finished product taster of the study will be the selected semi trained panelist. The taster will rate each siomai in a qualitative description of Like Extremely (9), Like Very Much (8), Like Moderately (7), Like Slightly (6), Neither Like Nor Dislike (5), Dislike Slightly (4), Dislike moderately (3), Dislike very much (2) and Dislike Extremely (1).

Data to be collected will be analyzed using the Analysis of Variance (ANOVA). Differences in treatment means will be compared using Least Significant Difference (LSD). The result will be interpreted at five percent level of significance.

Determination of Siomai Shelf Life and Nutritional Value

To determine the shelf life of the product, samples will be taken and observed under room temperature until the undesirable odor, color and molds will develop. To get nutritional value, samples will also be submitted in the DOST for laboratory analysis.

(14) Major Activities/Work plan

RESEARCH ACTIVITIES	Jan	Feb	Mar	Apr	May	Jun
Acquisition of materials						
Data Gathering and Analysis						
Technical reports/presentations						
UM/ Patent Application						
IEC materials/Flyers production						

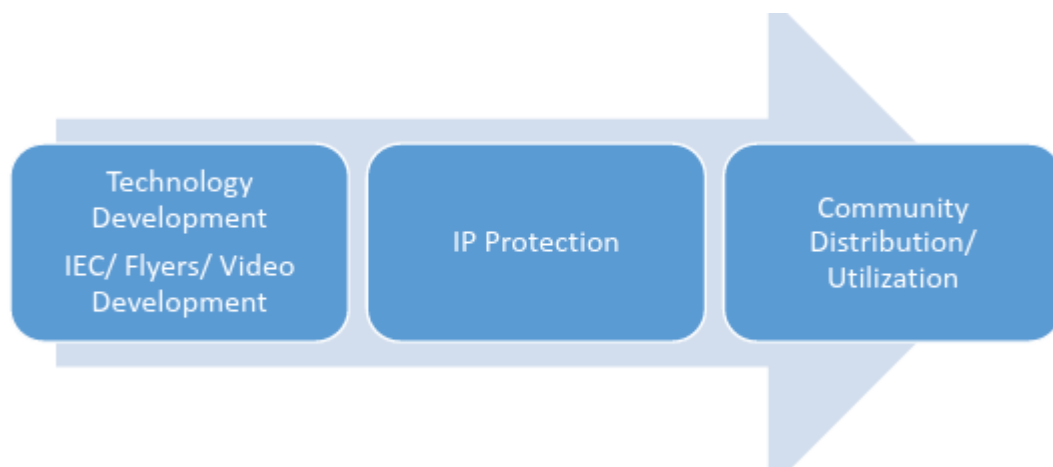
(15) Expected Output

EXPECTED OUTPUTS	DETAILS	TARGET
People Services	Train/ mentored native chicken growers and bamboo growers group or association for the utilization of DNC and Bamboo for siomai making.	100

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Publications	Develop/ produce Information, Education and Communication (IEC) and promotional materials for utilization of DNC, bamboo and malunggay for siomai production.	2	
Patents/IP	File 2 copyrights for the developed IEC materials and videos.	2	
Places and Partnerships	Forge linkages and partnerships to LGU's, Capiz Darag Growers Association, Capiz Tourism Office, DTI, and other organizations.	5	
Products	Healthy siomai.		
Policy	Develop one policy inputs for utilization of DNC, bamboo and malunggay siomai.	1	
Impact to Social/ Economics	<i>Social Impact:</i> • Healthy siomai snacks • Technology transfer and commercialization strengthened • Strengthened Public-Private Partnerships • Innovation ecosystem improved • Technopreneurship empowered • Local agribusinesses supported <i>Economic Impact:</i> • Increase income of native chicken, bamboo and malunggay growers		

(15) Technology Transfer Activity Design



(16) Targets Beneficiaries

- Darag Native Chicken Raisers
- Bamboo Growers
- Malungay Growers
- Entrepreneurs
- Nutritionists
- Parents
- Students

(17) Personnel Requirement

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Name	Role (Program,Project, Study Leader)	Field of Specialization
Dr. Maryneth B. Barrios	Study Leader	Animal Science
RA	Research Assistant	Food Expert

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (CapSU)
A. <i>Personnel Services (For studies needed the Research Assistant, informant/s, guide)</i>	1 @ 11,068 x 6 months	66,408.00
B. <i>Maintenance and Other Operating Expenses (MOOE)</i>		
Transportation allowance		10,000
Representation Expenses		20,000
Training Expenses		30,000
Supplies (DNC and other ingredients)		20,000
Communication		5,000
Laboratory Analysis		20,000
Siomai Steamer		5,000
Pressure cooker		10,000
C. <i>Capital Outlay</i>		
Meat Grinder		50,000
Total		236, 408.00

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